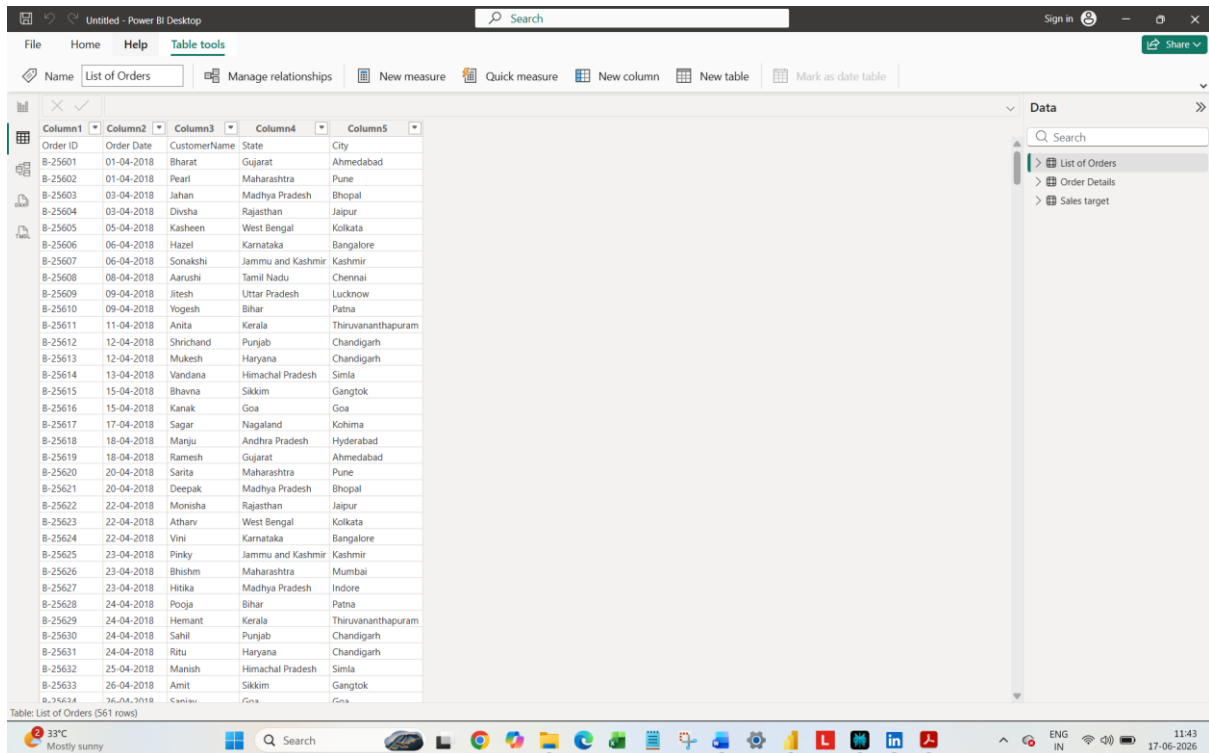


ScreenShots of Data Transformation & Data Modeling

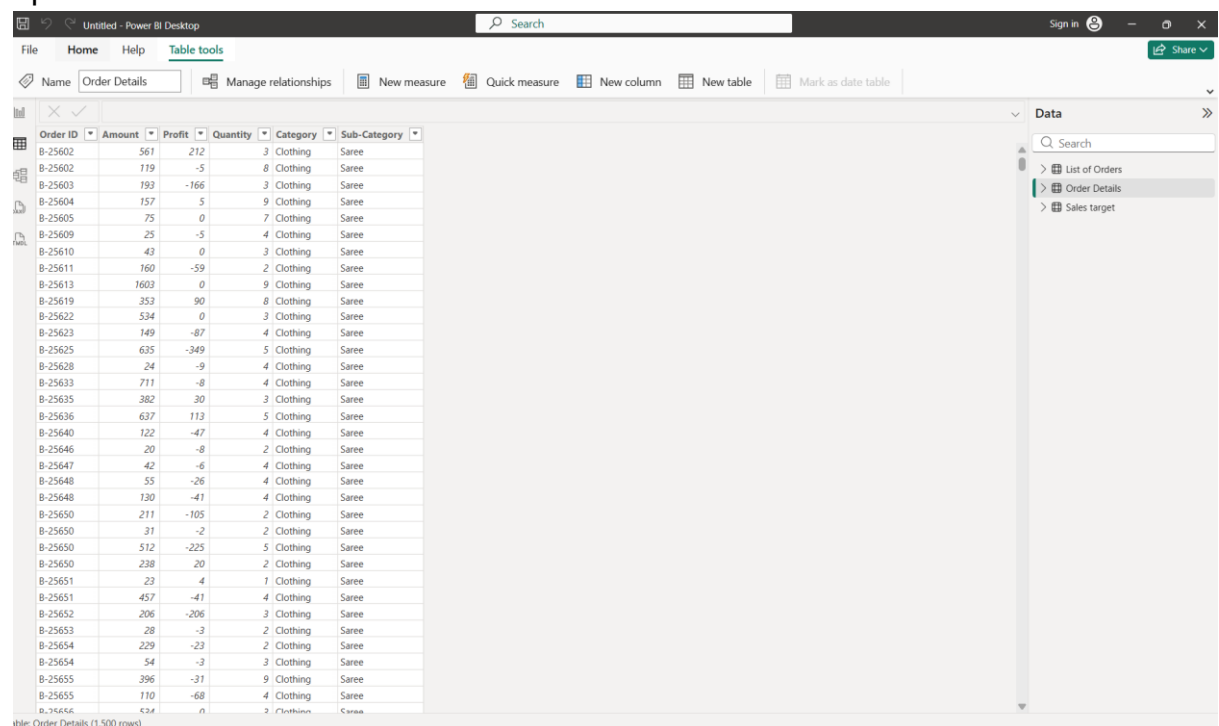
1. Data Import:

1) Import *List of Orders.csv* into Power BI.



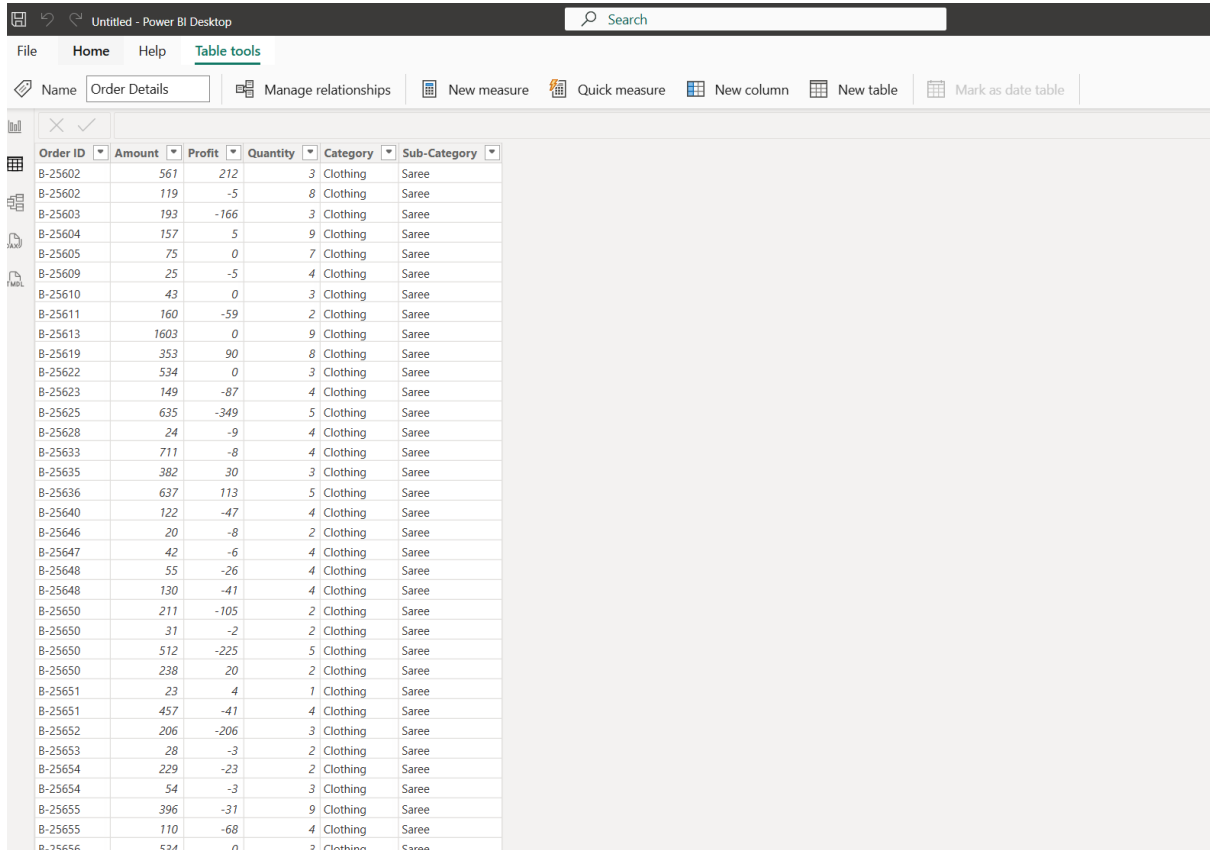
Column1	Column2	Column3	Column4	Column5
Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Itesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavia	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
B-25620	20-04-2018	Sarita	Maharashtra	Pune
B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
B-25623	22-04-2018	Atharv	West Bengal	Kolkata
B-25624	22-04-2018	Vini	Karnataka	Bangalore
B-25625	23-04-2018	Pinky	Jammu and Kashmir	Kashmir
B-25626	23-04-2018	Bhishm	Maharashtra	Mumbai
B-25627	23-04-2018	Hitika	Madhya Pradesh	Indore
B-25628	24-04-2018	Pooja	Bihar	Patna
B-25629	24-04-2018	Hemant	Kerala	Thiruvananthapuram
B-25630	24-04-2018	Sahil	Punjab	Chandigarh
B-25631	24-04-2018	Ritu	Haryana	Chandigarh
B-25632	25-04-2018	Manish	Himachal Pradesh	Simla
B-25633	26-04-2018	Amit	Sikkim	Gangtok
B-25634	26-04-2018	Sandhu	Goa	Goa

2) Open *the list of orders* in the **Power Query Editor** using the **Transform Data** option.



Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25602	561	212	3	Clothing	Saree
B-25602	119	-5	8	Clothing	Saree
B-25603	193	-166	3	Clothing	Saree
B-25604	157	5	9	Clothing	Saree
B-25605	75	0	7	Clothing	Saree
B-25609	25	-5	4	Clothing	Saree
B-25610	43	0	3	Clothing	Saree
B-25611	160	-59	2	Clothing	Saree
B-25613	1603	0	9	Clothing	Saree
B-25619	353	90	8	Clothing	Saree
B-25622	534	0	3	Clothing	Saree
B-25623	149	-87	4	Clothing	Saree
B-25625	635	-349	5	Clothing	Saree
B-25628	24	-9	4	Clothing	Saree
B-25633	711	-8	4	Clothing	Saree
B-25635	382	30	3	Clothing	Saree
B-25636	637	113	5	Clothing	Saree
B-25640	122	-47	4	Clothing	Saree
B-25646	20	-8	2	Clothing	Saree
B-25647	42	-6	4	Clothing	Saree
B-25648	55	-26	4	Clothing	Saree
B-25648	130	-41	4	Clothing	Saree
B-25650	211	-105	2	Clothing	Saree
B-25650	31	-2	2	Clothing	Saree
B-25650	512	-225	5	Clothing	Saree
B-25650	238	20	2	Clothing	Saree
B-25651	23	4	1	Clothing	Saree
B-25651	457	-41	4	Clothing	Saree
B-25652	206	-206	3	Clothing	Saree
B-25653	28	-3	2	Clothing	Saree
B-25654	229	-23	2	Clothing	Saree
B-25654	54	-3	3	Clothing	Saree
B-25655	396	-31	9	Clothing	Saree
B-25655	110	-68	4	Clothing	Saree
B-25656	634	0	3	Clothing	Saree

3) Import *Order Details.csv* and *Sales Target.csv* into Power Query Editor.



Power BI Desktop interface showing the Power Query Editor. The table 'Order Details' is loaded, displaying columns: Order ID, Amount, Profit, Quantity, Category, and Sub-Category. The data is as follows:

Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25602	561	212	3	Clothing	Saree
B-25602	119	-5	8	Clothing	Saree
B-25603	193	-166	3	Clothing	Saree
B-25604	157	5	9	Clothing	Saree
B-25605	75	0	7	Clothing	Saree
B-25609	25	-5	4	Clothing	Saree
B-25610	43	0	3	Clothing	Saree
B-25611	160	-59	2	Clothing	Saree
B-25613	1603	0	9	Clothing	Saree
B-25619	353	90	8	Clothing	Saree
B-25622	534	0	3	Clothing	Saree
B-25623	149	-87	4	Clothing	Saree
B-25625	635	-349	5	Clothing	Saree
B-25628	24	-9	4	Clothing	Saree
B-25633	711	-8	4	Clothing	Saree
B-25635	382	30	3	Clothing	Saree
B-25636	637	113	5	Clothing	Saree
B-25640	122	-47	4	Clothing	Saree
B-25646	20	-8	2	Clothing	Saree
B-25647	42	-6	4	Clothing	Saree
B-25648	55	-26	4	Clothing	Saree
B-25648	130	-41	4	Clothing	Saree
B-25650	211	-105	2	Clothing	Saree
B-25650	31	-2	2	Clothing	Saree
B-25650	512	-225	5	Clothing	Saree
B-25650	238	20	2	Clothing	Saree
B-25651	23	4	1	Clothing	Saree
B-25651	457	-41	4	Clothing	Saree
B-25652	206	-206	3	Clothing	Saree
B-25653	28	-3	2	Clothing	Saree
B-25654	229	-23	2	Clothing	Saree
B-25654	54	-3	3	Clothing	Saree
B-25655	396	-31	9	Clothing	Saree
B-25655	110	-68	4	Clothing	Saree
B-25656	524	0	2	Clothing	Saree

2. Row Limiting & Data Types:

- 1) Restrict the *List of Orders* table to the **first 500 rows**.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Data Type: Text • Replace Values • Unpivot Columns • Split Column • Merge Columns • Statistics • Trigonometry • Date • Time • Duration • Run R script • Run Python script

Queries [3] List of Orders Order Details Sales target

Table

fx = Table.SelectRows(*Trimmed Text", each [Order ID] <> null and [Order ID] <> "")

Order ID	Order Date	CustomerName	State	City
1	01-04-2018	Bharat	Gujarat	Ahmedabad
2	01-04-2018	Pearl	Maharashtra	Pune
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	03-04-2018	Divsha	Rajasthan	Jaipur
5	05-04-2018	Kasheen	West Bengal	Kolkata
6	06-04-2018	Hazel	Karnataka	Bangalore
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	09-04-2018	Yogesh	Bihar	Patna
11	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	12-04-2018	Shrichand	Punjab	Chandigarh
13	12-04-2018	Mukesh	Haryana	Chandigarh
14	13-04-2018	Vandana	Himachal Pradesh	Simla
15	15-04-2018	Bhavna	Sikkim	Gangtok
16	15-04-2018	Kanak	Goa	Goa
17	17-04-2018	Sagar	Nagaland	Kohima
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	20-04-2018	Sarita	Maharashtra	Pune
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	22-04-2018	Monisha	Rajasthan	Jaipur
23	22-04-2018	Atharv	West Bengal	Kolkata
24	22-04-2018	Vini	Karnataka	Bangalore
25	23-04-2018	Pinky	Jammu and Kashmir	Kashmir
26	23-04-2018	Bhoshm	Maharashtra	Mumbai
27	23-04-2018	Hitika	Madhya Pradesh	Indore
28	24-04-2018	Pooja	Bihar	Patna
29	24-04-2018	Hemant	Kerala	Thiruvananthapuram
30	24-04-2018	Sahil	Punjab	Chandigarh

5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 11:54

Query Settings

PROPERTIES Name List of Orders All Properties

APPLIED STEPS Source Promoted Headers Changed Type Cleaned Text Trimmed Text Filtered Rows

2) Set the **Order Date** column to the **Date** data type.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Data Type: Text • Replace Values • Unpivot Columns • Split Column • Merge Columns • Statistics • Trigonometry • Date • Time • Duration • Run R script • Run Python script

Queries [3] List of Orders Order Details Sales target

Table

fx = Table.SelectRows(*Trimmed Text", each [Order ID] <> null and [Order ID] <> "")

Order ID	Order Date	CustomerName	State	City
1	01-04-2018	Bharat	Gujarat	Ahmedabad
2	01-04-2018	Pearl	Maharashtra	Pune
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	03-04-2018	Divsha	Rajasthan	Jaipur
5	05-04-2018	Kasheen	West Bengal	Kolkata
6	06-04-2018	Hazel	Karnataka	Bangalore
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	09-04-2018	Yogesh	Bihar	Patna
11	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	12-04-2018	Shrichand	Punjab	Chandigarh
13	12-04-2018	Mukesh	Haryana	Chandigarh
14	13-04-2018	Vandana	Himachal Pradesh	Simla
15	15-04-2018	Bhavna	Sikkim	Gangtok
16	15-04-2018	Kanak	Goa	Goa
17	17-04-2018	Sagar	Nagaland	Kohima
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	20-04-2018	Sarita	Maharashtra	Pune
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	22-04-2018	Monisha	Rajasthan	Jaipur
23	22-04-2018	Atharv	West Bengal	Kolkata
24	22-04-2018	Vini	Karnataka	Bangalore
25	23-04-2018	Pinky	Jammu and Kashmir	Kashmir
26	23-04-2018	Bhoshm	Maharashtra	Mumbai
27	23-04-2018	Hitika	Madhya Pradesh	Indore
28	24-04-2018	Pooja	Bihar	Patna
29	24-04-2018	Hemant	Kerala	Thiruvananthapuram
30	24-04-2018	Sahil	Punjab	Chandigarh

Query Settings

PROPERTIES Name List of Orders All Properties

APPLIED STEPS Source Promoted Headers Changed Type Cleaned Text Trimmed Text Filtered Rows

3) Change the **Amount** and **Target** columns to **fixed decimal numbers**.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Transpose Reverse Rows Count Rows
Use First Row as Headers

Data Type: Fixed decimal number
Detect Data Type
Rename

Replace Values
Fill
Pivot Column
Unpivot Columns
Convert to List

Split Column
Format
Merge Columns
Extract
Parse

Statistics Standard Scientific
Trigonometry
Rounding
Information

Date Time Duration
Run R script
Run Python script

Queries [3]
List of Orders
Order Details
Sales target

Table

fx = Table.TransformColumnTypes(#"Filtered Rows",{{"Amount", Currency.Type}})

	Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	B-25601	1,275.00	-1148	7	Furniture	Bookcases
2	B-25601	66.00	-12	5	Clothing	Stole
3	B-25601	8.00	-2	3	Clothing	Hankierchief
4	B-25601	80.00	-56	4	Electronics	Electronic Games
5	B-25602	168.00	-111	2	Electronics	Phones
6	B-25602	424.00	-272	5	Electronics	Phones
7	B-25602	2,617.00	1151	4	Electronics	Phones
8	B-25602	561.00	212	3	Clothing	Saree
9	B-25602	119.00	-5	8	Clothing	Saree
10	B-25603	1,355.00	-60	5	Clothing	Trousers
11	B-25603	24.00	-30	1	Furniture	Chairs
12	B-25603	193.00	-166	3	Clothing	Saree
13	B-25603	180.00	5	3	Clothing	Trousers
14	B-25603	116.00	16	4	Clothing	Stole
15	B-25603	107.00	36	6	Clothing	Stole
16	B-25603	12.00	1	2	Clothing	Hankierchief
17	B-25603	38.00	18	1	Clothing	Kurti
18	B-25604	65.00	17	2	Clothing	T-shirt
19	B-25604	157.00	5	9	Clothing	Saree
20	B-25605	75.00	0	7	Clothing	Saree
21	B-25606	87.00	4	2	Clothing	Shirt
22	B-25607	50.00	15	4	Clothing	Leggings
23	B-25608	1,364.00	-1854	5	Furniture	Tables
24	B-25608	476.00	0	3	Furniture	Chairs
25	B-25608	257.00	23	5	Clothing	Hankierchief
26	B-25608	856.00	885	6	Electronics	Printers
27	B-25609	483.00	29	4	Electronics	Electronic Games
28	B-25609	25.00	-5	4	Clothing	Saree
29	B-25610	1,076.00	-38	4	Electronics	Printers
30	B-25610	107.00	-54	4	Clothing	Stole

Query Settings

PROPERTIES
Name
Order Details

APPLIED STEPS
Source
Promoted headers
Changed column type
Cleaned Text
Trimmed Text
Filtered Rows
X Changed Type

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Transpose Reverse Rows Count Rows
Use First Row as Headers

Data Type: Fixed decimal number
Detect Data Type
Rename

Replace Values
Fill
Pivot Column
Unpivot Columns
Convert to List

Split Column
Format
Merge Columns
Extract
Parse

Statistics Standard Scientific
Trigonometry
Rounding
Information

Date Time Duration
Run R script
Run Python script

Queries [3]
List of Orders
Order Details
Sales target

Table

fx = Table.TransformColumnTypes(#"Trimmed Text",{{"Target", Currency.Type}})

	Month of Order Date	Category	Target
1	18-04-2026	Furniture	10,400.00
2	18-05-2026	Furniture	10,500.00
3	18-06-2026	Furniture	10,600.00
4	18-07-2026	Furniture	10,800.00
5	18-08-2026	Furniture	10,900.00
6	18-09-2026	Furniture	11,000.00
7	18-10-2026	Furniture	11,100.00
8	18-11-2026	Furniture	11,300.00
9	18-12-2026	Furniture	11,400.00
10	19-01-2026	Furniture	11,500.00
11	19-02-2026	Furniture	11,600.00
12	19-03-2026	Furniture	11,800.00
13	18-04-2026	Clothing	12,000.00
14	18-05-2026	Clothing	12,000.00
15	18-06-2026	Clothing	12,000.00
16	18-07-2026	Clothing	14,000.00
17	18-08-2026	Clothing	14,000.00
18	18-09-2026	Clothing	14,000.00
19	18-10-2026	Clothing	16,000.00
20	18-11-2026	Clothing	16,000.00
21	18-12-2026	Clothing	16,000.00
22	19-01-2026	Clothing	16,000.00
23	19-02-2026	Clothing	16,000.00
24	19-03-2026	Clothing	16,000.00
25	18-04-2026	Electronics	9,000.00
26	18-05-2026	Electronics	9,000.00
27	18-06-2026	Electronics	9,000.00
28	18-07-2026	Electronics	9,000.00
29	18-08-2026	Electronics	9,000.00
30	18-09-2026	Electronics	9,000.00

Query Settings

PROPERTIES
Name
Sales target

APPLIED STEPS
Source
Promoted headers
Changed column type
Filtered Rows
Cleaned Text
Trimmed Text
X Changed Type

3. Text Formatting:

- 1) Format the **CustomerName** column to **proper case** to ensure consistent capitalization.

Untitled - Power Query Editor

Home Transform Add Column View Tools Help

Table

Series [3]

Table.TransformColumns(#"Trimmed Text3",{"CustomerName", Text.Proper, type text})

Order ID	Order Date	CustomerName	State	City
1	01-04-2018	Bharat	Gujarat	Ahmedabad
2	01-04-2018	Pearl	Maharashtra	Pune
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	03-04-2018	Divsha	Rajasthan	Jaipur
5	05-04-2018	Kasheen	West Bengal	Kolkata
6	06-04-2018	Hazel	Karnataka	Bangalore
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	09-04-2018	Yogesh	Bihar	Patna
11	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	12-04-2018	Shrichand	Punjab	Chandigarh
13	12-04-2018	Mukesh	Haryana	Chandigarh
14	13-04-2018	Vandana	Himachal Pradesh	Simla
15	15-04-2018	Bhavna	Sikkim	Gangtok
16	15-04-2018	Kanak	Goa	Goa
17	17-04-2018	Sagar	Nagaland	Kohima
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	20-04-2018	Sarita	Maharashtra	Pune
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	22-04-2018	Monisha	Rajasthan	Jaipur
23	22-04-2018	Atharv	West Bengal	Kolkata
24	22-04-2018	Vini	Karnataka	Bangalore

Query Settings

PROPERTIES

Name

List of Orders

APPLIED STEPS

Source

Promoted Headers

Changed Type

Cleaned Text

Trimmed Text

Filtered Rows

Changed Type1

Filtered Rows1

Cleaned Text1

Trimmed Text1

Cleaned Text2

Trimmed Text2

Cleaned Text3

Trimmed Text3

Capitalized Each Word

4. Column Creation:

1) Merge the City and State columns to create a new column named Location in the format: **City, State**

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Table

Series [6]

Table.AddColumn(#"Changed Type1", "Location", each Text.Combine([City], [State], ", "), type text)

Order ID	Order Date	CustomerName	State	City	Location
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
2	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
4	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan
5	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal
6	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh
10	09-04-2018	Yogesh	Bihar	Patna	Patna,Bihar
11	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala
12	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh,Punjab
13	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana
14	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh
15	15-04-2018	Bhavna	Sikkim	Gangtok	Gangtok,Sikkim
16	15-04-2018	Kanak	Goa	Goa	Goa,Goa
17	17-04-2018	Sagar	Nagaland	Kohima	Kohima,Nagaland
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh
19	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat
20	20-04-2018	Sarita	Maharashtra	Pune	Pune,Maharashtra
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
22	22-04-2018	Monisha	Rajasthan	Jaipur	Jaipur,Rajasthan
23	22-04-2018	Atharv	West Bengal	Kolkata	Kolkata,West Bengal
24	22-04-2018	Vini	Karnataka	Bangalore	Bangalore,Karnataka
25	23-04-2018	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
26	23-04-2018	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra
27	23-04-2018	Hittika	Madhya Pradesh	Indore	Indore,Madhya Pradesh
28	24-04-2018	Pooja	Bihar	Patna	Patna,Bihar
29	24-04-2018	Hemant	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala
30	24-04-2018	Sahil	Punjab	Chandigarh	Chandigarh,Punjab

Query Settings

PROPERTIES

Name

List of Orders

APPLIED STEPS

Source

Promoted Headers

Changed Type

Cleaned Text

Trimmed Text

Filtered Rows

Changed Type1

Filtered Rows1

Cleaned Text1

Trimmed Text1

Cleaned Text2

Trimmed Text2

Cleaned Text3

Trimmed Text3

Changed Type1

Inserted Merged Column

2) Create a custom column named **Profit Margin** calculated as: **Profit / Amount** (expressed as a percentage)

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Enter Data * Data source settings * Parameters * Output Data * Query * Properties * Advanced Editor * Manage * Choose Columns * Remove Columns * Keep Rows * Remove Rows * Sort * Data Type: Percentage * Merge Queries * Append Queries * Combine Files * Combine

Queries [6]

List of Orders

Order Details

Sales target

Orders Data

Sales target Duplicate

Order Details Duplicate

fx = Table.TransformColumnTypes(*Added Custom*,{"Profit Margin", Percentage.Type})

	\$ Amount	% Profit	% Quantity	% Category	% Sub-Category	% Profit Margin
1	1,275.00	-1148	7	Furniture	Bookcases	-90.04%
2	66.00	-12	5	Clothing	Stole	-18.18%
3	8.00	-2	3	Clothing	Hankerchief	-25.00%
4	80.00	-56	4	Electronics	Electronic Games	-70.00%
5	168.00	-111	2	Electronics	Phones	-66.07%
6	424.00	-272	5	Electronics	Phones	-64.15%
7	2,617.00	1151	4	Electronics	Phones	43.98%
8	561.00	212	3	Clothing	Saree	37.79%
9	119.00	-5	8	Clothing	Saree	-4.20%
10	1,355.00	-60	5	Clothing	Trousers	-4.43%
11	24.00	-30	1	Furniture	Chairs	-125.00%
12	193.00	-166	3	Clothing	Saree	-86.01%
13	180.00	5	3	Clothing	Trousers	2.78%
14	116.00	16	4	Clothing	Stole	13.79%
15	107.00	36	6	Clothing	Stole	33.64%
16	12.00	1	2	Clothing	Hankerchief	8.33%
17	38.00	18	1	Clothing	Kurti	47.37%
18	65.00	17	2	Clothing	T-shirt	26.15%
19	157.00	5	9	Clothing	Saree	3.18%
20	75.00	0	7	Clothing	Saree	0.00%
21	87.00	4	2	Clothing	Shirt	4.60%
22	50.00	15	4	Clothing	Leggings	30.00%
23	1,364.00	-1864	5	Furniture	Tables	-136.66%
24	476.00	0	3	Furniture	Chairs	0.00%
25	257.00	23	5	Clothing	Hankerchief	8.95%
26	856.00	385	6	Electronics	Printers	44.98%
27	485.00	29	4	Electronics	Electronic Games	5.98%
28	25.00	-5	4	Clothing	Saree	-20.00%
29	1,076.00	-38	4	Electronics	Printers	-3.53%

Query Settings

PROPERTIES

Name

Order Details

APPLIED STEPS

Source

Promoted headers

Changed column type

Filtered Rows1

Cleaned Text

Trimmed Text

Cleaned Text1

Trimmed Text1

Changed Type

Added Custom

Changed Type1

Added Conditional Column

5. Conditional Column:

1)Add a conditional column named **Profit Status** based on: ■ Profit < 0 → *Loss* ■ Profit = 0 → *Break-Even*

■ Profit > 0 → *Profit*

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Enter Data * Data source settings * Parameters * Output Data * Query * Properties * Advanced Editor * Manage * Choose Columns * Remove Columns * Keep Rows * Remove Rows * Sort * Data Type: Percentage * Merge Queries * Append Queries * Combine Files * Combine

Queries [6]

List of Orders

Order Details

Sales target

Orders Data

Sales target Duplicate

Order Details Duplicate

fx = Table.AddColumn(*Changed Type1*, "Profit Status", each if [Profit] < 0 then "Loss" else if [Profit] = 0 then "Break-Even" else if

	\$ Amount	% Profit	% Quantity	% Category	% Sub-Category	% Profit Margin
1	1,275.00	-1148	7	Furniture	Bookcases	-90.04%
2	66.00	-12	5	Clothing	Stole	-18.18%
3	8.00	-2	3	Clothing	Hankerchief	-25.00%
4	80.00	-56	4	Electronics	Electronic Games	-70.00%
5	168.00	-111	2	Electronics	Phones	-66.07%
6	424.00	-272	5	Electronics	Phones	-64.15%
7	2,617.00	1151	4	Electronics	Phones	43.98%
8	561.00	212	3	Clothing	Saree	37.79%
9	119.00	-5	8	Clothing	Saree	-4.20%
10	1,355.00	-60	5	Clothing	Trousers	-4.43%
11	24.00	-30	1	Furniture	Chairs	-125.00%
12	193.00	-166	3	Clothing	Saree	-86.01%
13	180.00	5	3	Clothing	Trousers	2.78%
14	116.00	16	4	Clothing	Stole	13.79%
15	107.00	36	6	Clothing	Stole	33.64%
16	12.00	1	2	Clothing	Hankerchief	8.33%
17	38.00	18	1	Clothing	Kurti	47.37%
18	65.00	17	2	Clothing	T-shirt	26.15%
19	157.00	5	9	Clothing	Saree	3.18%
20	75.00	0	7	Clothing	Saree	0.00%
21	87.00	4	2	Clothing	Shirt	4.60%
22	50.00	15	4	Clothing	Leggings	30.00%
23	1,364.00	-1864	5	Furniture	Tables	-136.66%
24	476.00	0	3	Furniture	Chairs	0.00%
25	257.00	23	5	Clothing	Hankerchief	8.95%
26	856.00	385	6	Electronics	Printers	44.98%
27	485.00	29	4	Electronics	Electronic Games	5.98%
28	25.00	-5	4	Clothing	Saree	-20.00%
29	1,076.00	-38	4	Electronics	Printers	-3.53%

Query Settings

PROPERTIES

Name

Order Details

APPLIED STEPS

Source

Promoted headers

Changed column type

Filtered Rows1

Cleaned Text

Trimmed Text

Cleaned Text1

Trimmed Text1

Changed Type

Added Custom

Changed Type1

Added Conditional Column

6. Merging Data (Joins)

- Merge the *List of Orders* and *Order Details* tables using **Order ID**.
- Name the merged output table as **Orders Data**.

The screenshot shows the Power BI Desktop interface with a table named 'Orders Data' loaded. The table has 30 rows and 7 columns: Order Date, CustomerName, State, City, Location, and Orders Data. The 'Orders Data' column contains the text 'Table' for each row. The 'Query Settings' pane on the right shows the 'Properties' tab with the name 'Orders Data' and the 'Applied Steps' pane showing 'Renamed Columns', 'Sorted Rows', and 'Filtered Rows'.

	Order Date	CustomerName	State	City	Location	Orders Data
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	Table
2	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra	Table
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Table
4	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan	Table
5	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal	Table
6	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka	Table
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	Table
8	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Table
9	09-04-2018	Ilitesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh	Table
10	09-04-2018	Yogesh	Bihar	Patna	Patna,Bihar	Table
11	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala	Table
12	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh,Punjab	Table
13	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana	Table
14	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh	Table
15	15-04-2018	Bhavna	Sikkim	Gangtok	Gangtok,Sikkim	Table
16	15-04-2018	Kanak	Goa	Goa	Goa,Goa	Table
17	17-04-2018	Sagar	Nagaland	Kohima	Kohima,Nagaland	Table
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh	Table
19	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat	Table
20	20-04-2018	Sarita	Maharashtra	Pune	Pune,Maharashtra	Table
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Table
22	22-04-2018	Monisha	Rajasthan	Jaipur	Jaipur,Rajasthan	Table
23	22-04-2018	Atharv	West Bengal	Kolkata	Kolkata,West Bengal	Table
24	22-04-2018	Vini	Karnataka	Bangalore	Bangalore,Karnataka	Table
25	23-04-2018	Pinkiy	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	Table
26	23-04-2018	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra	Table
27	23-04-2018	Hitika	Madhya Pradesh	Indore	Indore,Madhya Pradesh	Table
28	24-04-2018	Pooja	Bihar	Patna	Patna,Bihar	Table
29	24-04-2018	Hemant	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala	Table
30						

8. Sorting and Filtering Data

1. Sort records in the **Orders Data** table by **Order Date** in **descending order**.

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Data Sources Parameters Output Data Query Properties Advanced Editor Manage Columns * Remove Columns * Keep Rows * Remove Rows * Sort Split Column * Group By Data Type: Date * Use First Row as Headers * Merge Queries * Append Queries * Combine Files

Queries [6]

Table.Sort(#"Filtered Rows",{"Order Date", Order.Descending})

	Order Date	CustomerName	State	City	Location	Orders Data
1	31-03-2019	Hitesh	Madhya Pradesh	Indore	Indore,Madhya Pradesh	Table
2	30-03-2019	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra	Table
3	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	Table
4	28-03-2019	Vini	Karnataka	Bangalore	Bangalore,Karnataka	Table
5	28-03-2019	Atharv	West Bengal	Kolkata	Kolkata,West Bengal	Table
6	28-03-2019	Monisha	Rajasthan	Jaipur	Jaipur,Rajasthan	Table
7	27-03-2019	Deepak	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Table
8	27-03-2019	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh	Table
9	27-03-2019	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat	Table
10	27-03-2019	Sarita	Maharashtra	Pune	Pune,Maharashtra	Table
11	27-03-2019	Sagar	Nagaland	Kohima	Kohima,Nagaland	Table
12	26-03-2019	Bhavna	Sikkim	Gangtok	Gangtok,Sikkim	Table
13	26-03-2019	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana	Table
14	26-03-2019	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh	Table
15	26-03-2019	Shrichand	Punjab	Chandigarh	Chandigarh,Punjab	Table
16	26-03-2019	Kanak	Goa	Goa	Goa,Goa	Table
17	25-03-2019	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala	Table
18	24-03-2019	Yogesh	Bihar	Patna	Patna,Bihar	Table
19	23-03-2019	Jitesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh	Table
20	22-03-2019	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal	Table
21	22-03-2019	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan	Table
22	22-03-2019	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	Table
23	22-03-2019	Hazel	Karnataka	Bangalore	Bangalore,Karnataka	Table
24	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Table
25	21-03-2019	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	Table
26	21-03-2019	Pournamasi	Madhya Pradesh	Indore	Indore,Madhya Pradesh	Table
27	21-03-2019	Pearl	Maharashtra	Pune	Pune,Maharashtra	Table
28	21-03-2019	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Table
29	20-03-2019	Sweta	Maharashtra	Mumbai	Mumbai,Maharashtra	Table
30						

Query Settings

PROPERTIES

Name

Orders Data

All Properties

APPLIED STEPS

Source

Renamed Columns

Sorted Rows

Sorted Rows1

2. Apply filters to analyse orders from a specific state (e.g., Tamil Nadu).

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Data Sources Parameters Output Data Query Properties Advanced Editor Manage Columns * Remove Columns * Keep Rows * Remove Rows * Sort Split Column * Group By Data Type: Text * Use First Row as Headers * Merge Queries * Append Queries * Combine Files

Queries [6]

Table.SelectRows(#"Sorted Rows1", each ([State] = "Tamil Nadu"))

	Order Date	CustomerName	State	City	Location	Orders Data
1	22-02-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Table
2	14-02-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Table
3	09-02-2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Table
4	13-11-2018	Akshay	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Table
5	21-09-2018	Dinesh	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Table
6	11-07-2018	Surabhi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Table
7	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Table
8	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Table

Query Settings

PROPERTIES

Name

Orders Data

All Properties

APPLIED STEPS

Source

Renamed Columns

Sorted Rows

Filtered Rows

Sorted Rows1

Filtered Rows1

9. Grouping and Aggregating Data

- Duplicate the *Order Details* table and perform the following:
 - Count of Order ID
 - Average Profit by Category
 - Total Amount by Sub-Category

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Keep Remove Split Group Data Type: Text Merge Queries Append Queries Combine Files Combine

Queries [6]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Sales target Duplicate
- Order Details Duplicate

Table.Group(*Added Conditional Column*, {"Order ID", "Category", "Sub-Category"}, {"Order Count", each Table.RowCount(_), Int64.Type},

Order ID	Category	Sub-Category	Order Count	Average Profit	Total Amount
1 B-25601	Furniture	Bookcases	2	-1148	2275
2 B-25601	Clothing	Stole	2	-12	66
3 B-25601	Clothing	Hankerchief	2	-2	8
4 B-25601	Electronics	Electronic Games	2	-56	80
5 B-25602	Electronics	Phones	3	256	3209
6 B-25602	Clothing	Saree	2	103.5	680
7 B-25603	Clothing	Trousers	2	-27.5	1535
8 B-25603	Furniture	Chairs	2	-30	24
9 B-25603	Clothing	Saree	2	-166	193
10 B-25603	Clothing	Stole	2	26	223
11 B-25603	Clothing	Hankerchief	2	2	12
12 B-25603	Clothing	Kurti	2	18	38
13 B-25604	Clothing	T-shirt	2	17	65
14 B-25604	Clothing	Saree	2	5	157
15 B-25605	Clothing	Saree	2	0	75
16 B-25606	Clothing	Shirt	2	4	87
17 B-25607	Clothing	Leggings	2	15	50
18 B-25608	Furniture	Tables	2	-1864	1364
19 B-25608	Furniture	Chairs	2	0	476
20 B-25608	Clothing	Hankerchief	2	23	257
21 B-25608	Electronics	Printers	2	385	856
22 B-25609	Electronics	Electronic Games	2	29	485
23 B-25609	Clothing	Saree	2	-5	25
24 B-25610	Electronics	Printers	2	-316	1857
25 B-25610	Clothing	Stole	2	-54	107
26 B-25610	Electronics	Accessories	2	-55	68
27 B-25610	Clothing	Saree	2	0	43
28 B-25610	Furniture	Furnishings	2	-5	30
29 B-25611	Clothing	Saree	2	-59	160
30 B-25612	Furniture	Chairs	2	-55	259

Query Settings

PROPERTIES

Name

Order Details Duplicate

APPLIED STEPS

- Source
- Promoted headers
- Changed column type
- Filtered Rows
- Filtered Rows1
- Cleaned Text
- Trimmed Text
- Cleaned Text1
- Trimmed Text1
- Changed Type
- Added Custom
- Changed Type1
- Added Conditional Column
- Grouped Rows

- Duplicate the *Sales Target* table and aggregate the total Target by **Month of order date**.

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Keep Remove Split Group Data Type: Text Merge Queries Append Queries Combine Files Combine

Queries [6]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Sales target Duplicate
- Order Details Duplicate

Table.Group(*Changed Type*, {"Month"}, {"Total Target", each List.Sum([Target]), type nullable number})

Month	Total Target
1 April	31400
2 May	31500
3 June	31600
4 July	33800
5 August	33900
6 September	34000
7 October	36200
8 November	36300
9 December	36400
10 January	43500
11 February	43600
12 March	43800

Query Settings

PROPERTIES

Name

Sales target Duplicate

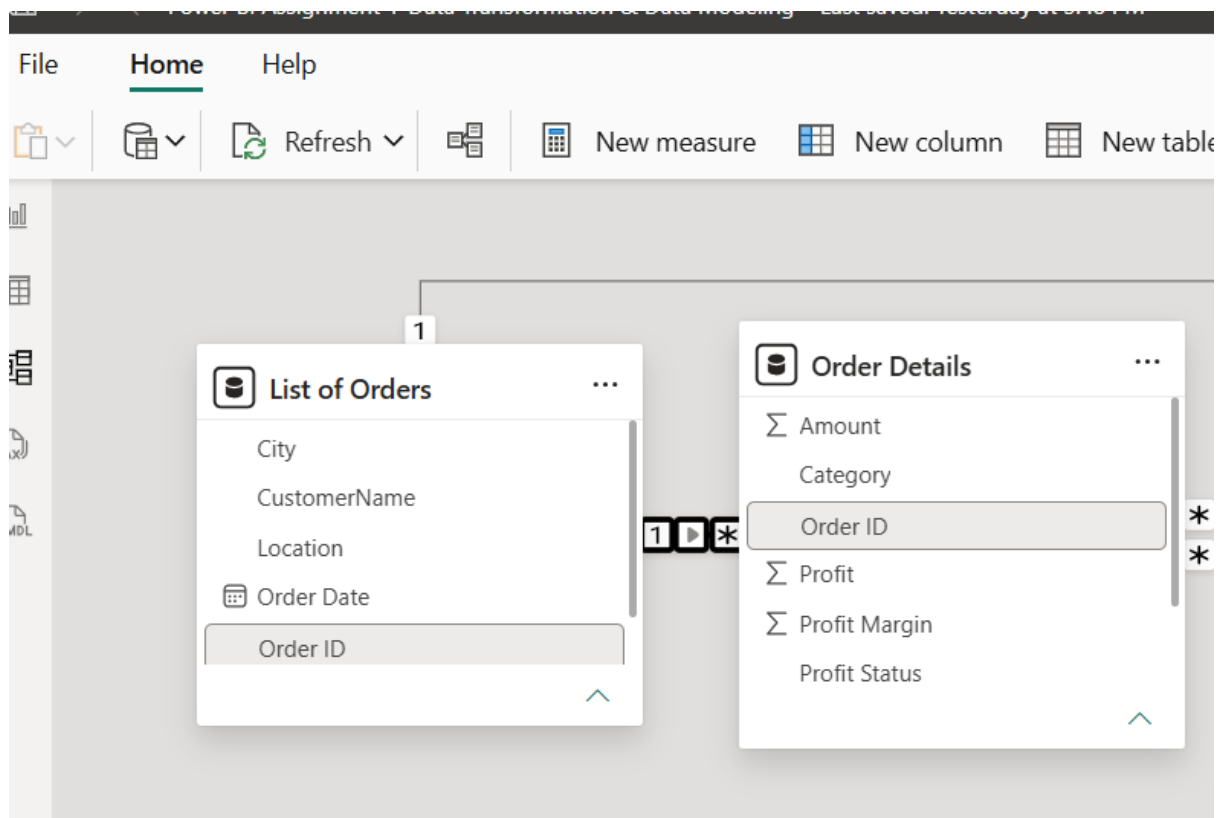
APPLIED STEPS

- Source
- Promoted headers
- Changed column type
- Filtered Rows
- Cleaned Text
- Cleaned Text1
- Duplicated Column
- Renamed Columns
- Extracted Month Name
- Changed Type
- Grouped Rows

10. Data Modeling:

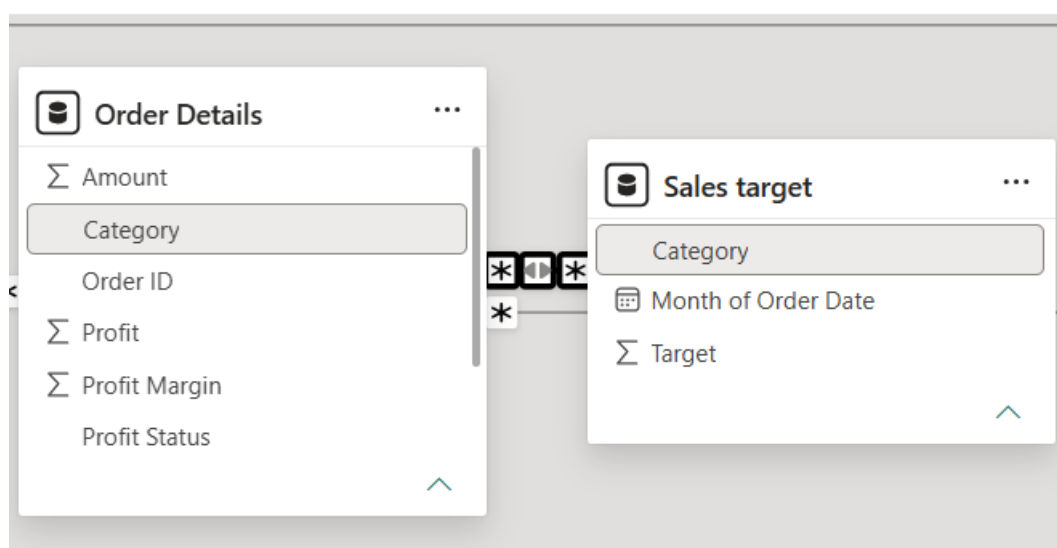
1) Establish a relationship between:

- List of Orders* and *Order Details* using **Order ID**



2) Establish a relationship between:

- *Order Details* and *Sales Target* using **Category**



- Use **Manage Relationships** to ensure relationships are active and correctly defined.

